

Longest Common Subsequence - DP

LCS $a = "a b c d e"$ $f(a, b, i, j)$

Recursion $b = "a c e"$

Logic: if $(a[i] == b[j])$ // match

return $1 + f(a, b, i+1, j+1)$ (next values)

else
return $\max(f(a, b, i+1, j), f(a, b, i, j+1))$

2D - DP

Algorithm

mxn

a	b	c	d	e
0	1	2	3	4
b	2			
d	1	1	1	
e	0	0		1

LCS
= 2

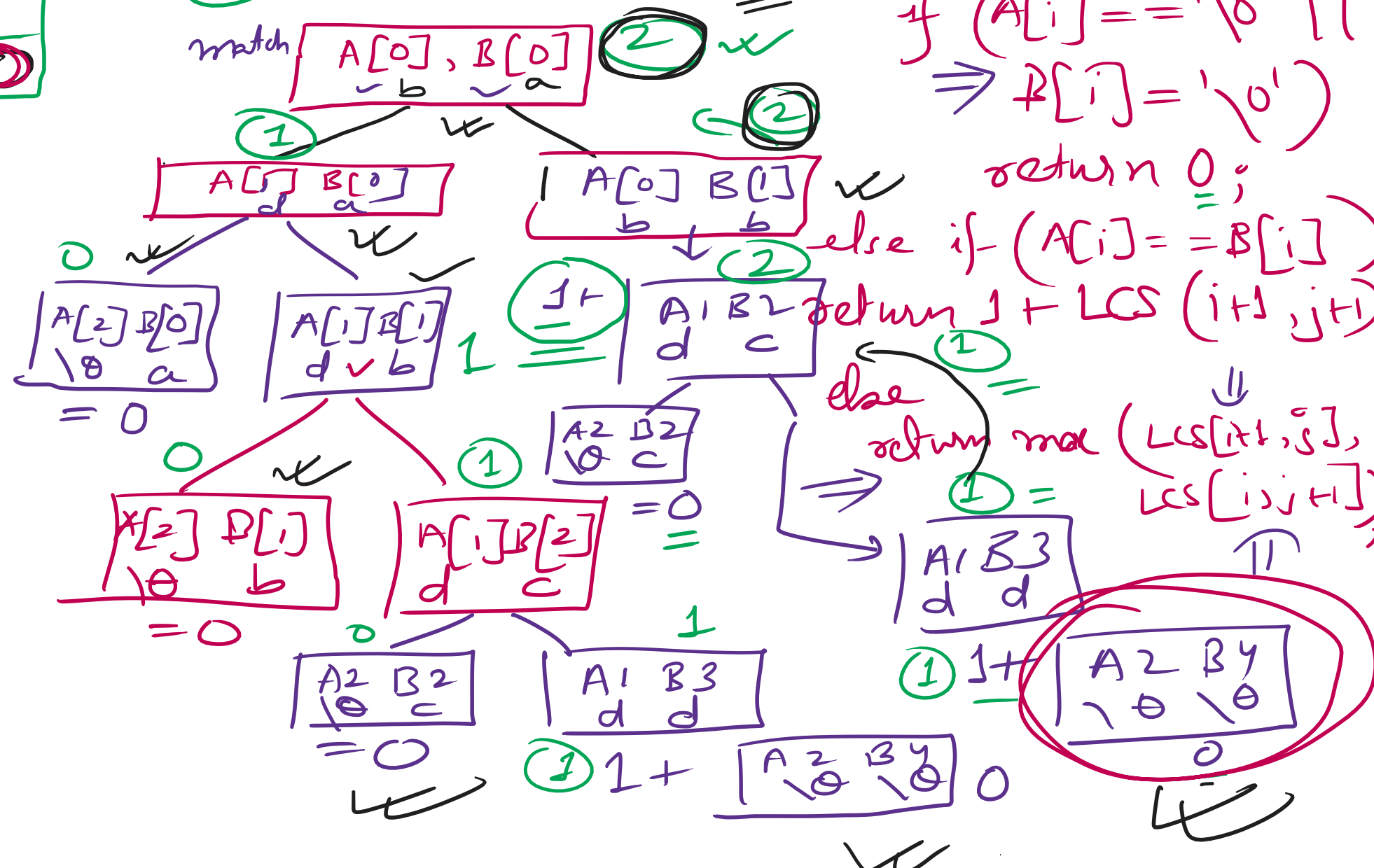
a	b	c	d	e
0	1	2	3	4
b	2			
d	1	1	1	
e	0	0		1

int LCS(i, j)

if $(A[i] == '0' || B[j] == '0')$
return 0;

else if $(A[i] == B[j])$
return $1 + \text{LCS}(i+1, j+1)$

else
return $\max(\text{LCS}(i+1, j), \text{LCS}(i, j+1))$



Memoization

$A = [b, d]$ ✓ 2

$B = [a, b, c, d]$ ✓ 4

0	0	0	0	0
1	0	1	1	1
2	0	0	1	1

LCS = [b, d]

if $(A[i] == B[j])$
LCS[i][j] = 1 + LCS[i-1][j-1]
length else

LCS[i][j] = max
(LCS[i-1][j], LCS[i][j-1])

[match → dia+1]
[unmatch → max row, col]

Example → 2 (one)

Str 1 = Stone - (5) → 6
Str 2 = longest - (7) → 8

DRY RUN
m → d+1
u → m pr pc

0	0	0	0	0	0	0	0
1	0	0	0	0	0	1	1
2	0	0	0	0	0	1	2
3	0	0	1	1	1	1	2
4	0	0	1	2	2	2	2
5	0	0	1	2	2	3	3

(i-1, j-1)

length

3

(b = length + 1) 0

* Thursday → Graphs (Test) **

Recursion + BT + DP → Thursday

1:00 to 3:00